VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF ELECTRONICS ENGINEERING

Course : Antenna and Microwave Engineering Lab (BECE305L)
Lab Slot : L23+L24
Faculty Name : YOGESH KUMAR CHOUKIKER

Task/ Experiment No. : 3	**Date : 30/09/2025**
Name of the Student : HARI KRUTHIK SK	**Regd. No. : 23BEC0358**

AIM:

- To design and simulate a 90° Branch-Line Coupler and a 180° Hybrid Coupler (Rat-Race Coupler).
- To determine their scattering parameters (S-Parameters), specifically observing the power division and phase difference between output ports.

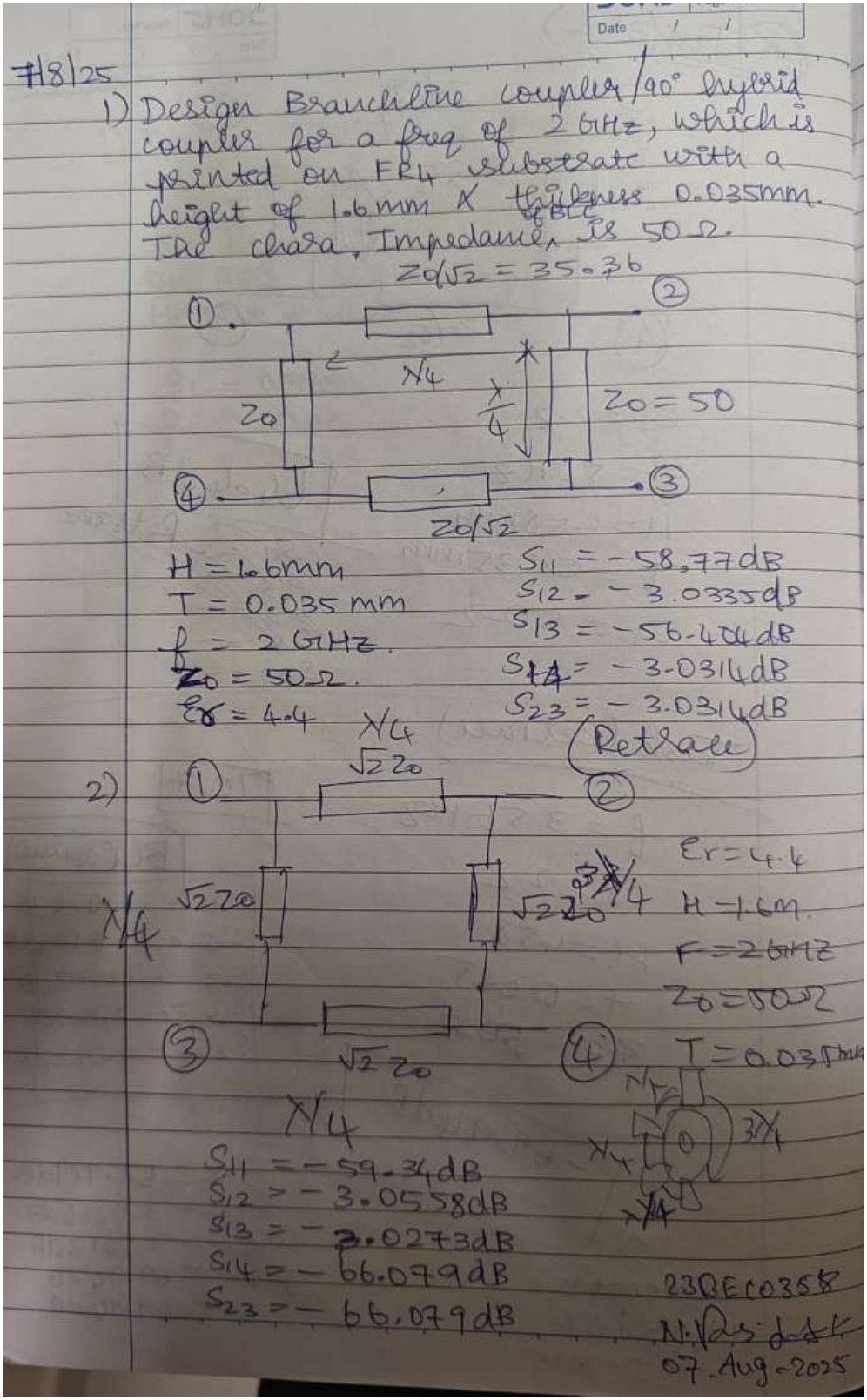
For Branchline Coupler :

- Zo = 50 ohm
- ε_r = 2.2
- H = 1.58 mm
- L = 0.02
- F = 5 GHz

For 180° Hybrid Coupler :

- Zo = 50 ohm
- ε_r = 2.2
- H = 1.58 mm
- L = 0.02
- F = 3.5 GHz

LAB TASK EXAMPLE QUESTIONS :



7/8/25

1) Design Branchline coupler 90° hybrid coupler for a freq of 2 GHz, which is printed on FR4 substrate with a height of 1.6mm & thickness 0.035mm. The chara. Impedance is 50Ω.

$Z_0/\sqrt{2} = 35.36$

H = 1.6mm
T = 0.035 mm
f = 2 GHz.
Z_0 = 50Ω.
ε_r = 4.4

$S_{11} = -58.77$dB
$S_{12} = -3.0335$dB
$S_{13} = -56.404$dB
$S_{14} = -3.0314$dB
$S_{23} = -3.0314$dB
(Retrace)

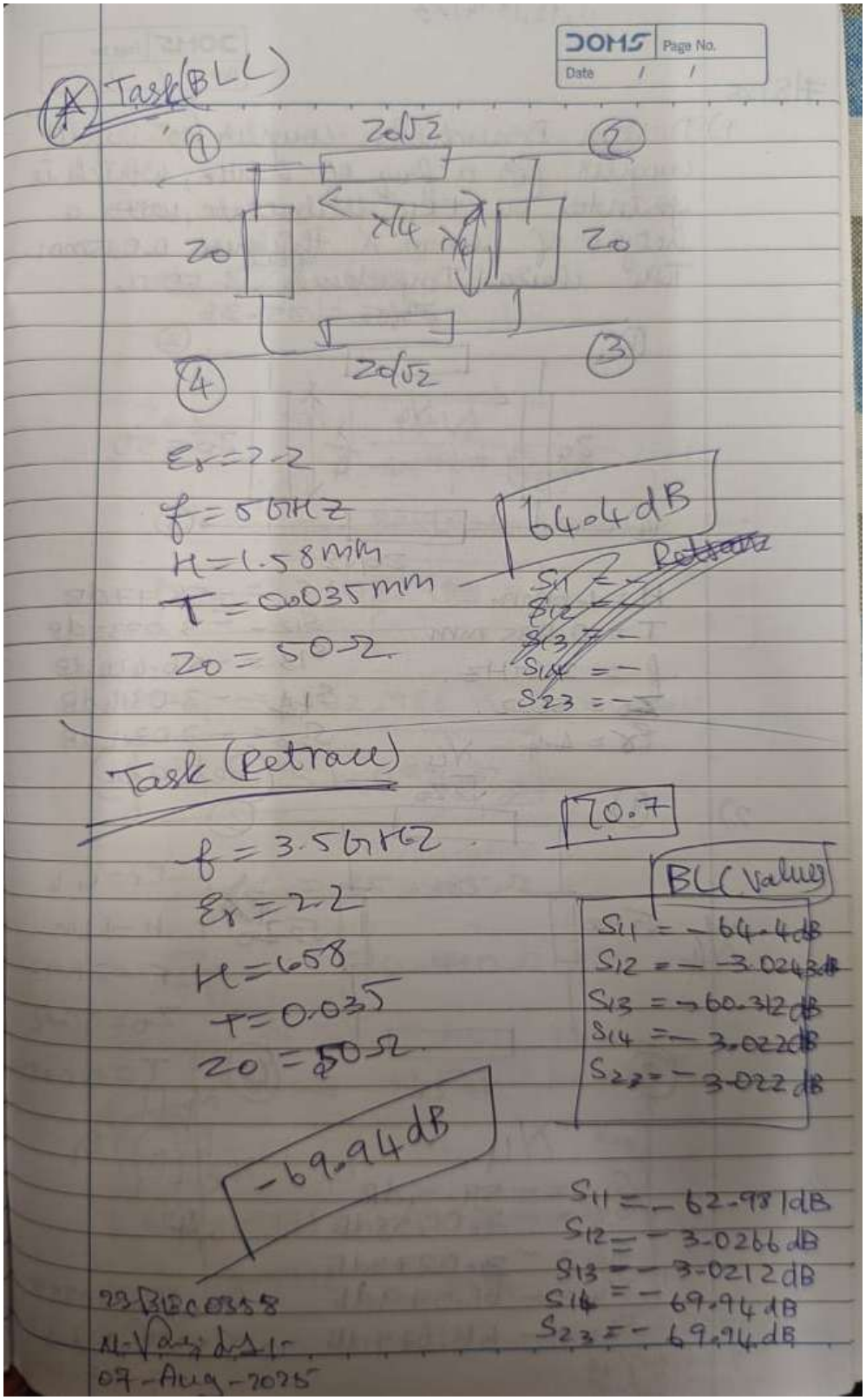
2)

$\varepsilon_r = 4.4$
H = 1.6m.
F = 2GHz
$Z_0 = 50\Omega$
T = 0.035mm

$S_{11} = -59.34$dB
$S_{12} = -3.0558$dB
$S_{13} = -3.0273$dB
$S_{14} = -66.079$dB
$S_{23} = -66.079$dB

23BEC0358
07 Aug 2025

TASK 3 QUESTIONS :



Expt. No.: Task-3

Date : 7/8/25

Branch Line Coupler

Objectives:

3. To design and simulation of a branch line coupler.
4. To determine the scattering parameters of branch line coupler.

Equipmentrequired: AWR Microwave Office software

Specifications:

Characteristic impedance	Z_0= 50 Ω
Operating frequency	f = 5 GHz
Substrate thickness	H= 1.58 mm
Dielectric constant	ε_r= 2.2
Loss tangent	L= 0.02

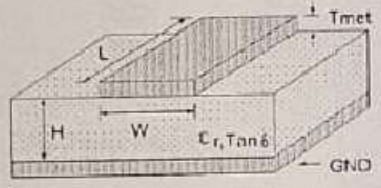
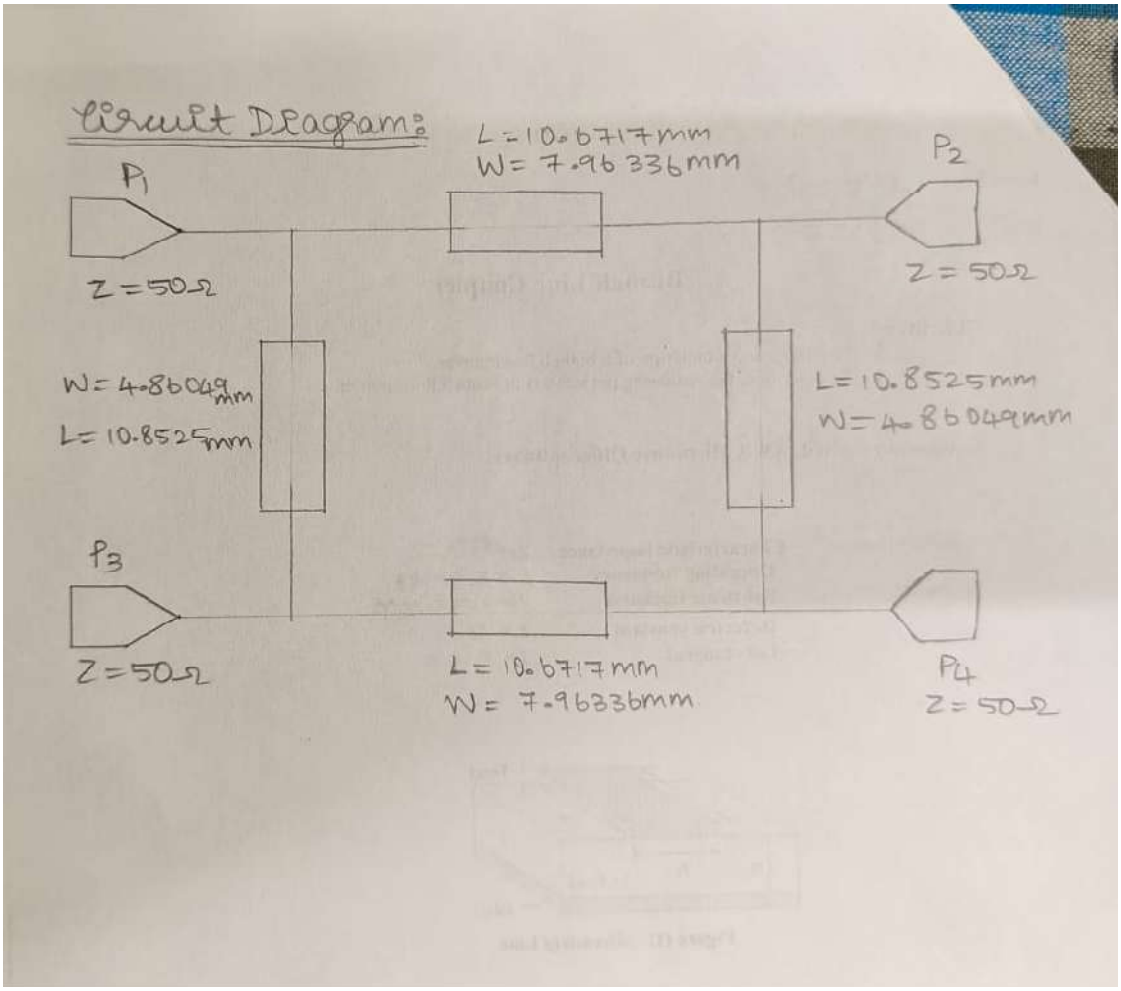


Figure (1) Microstrip Line

Theory :Branch line couplers are 3 dB directional couplers with a 90° phase difference in the outputs of the through and coupled ports.This type of hybrid is often made in microstrip line form as shown in figure 2.It is also known as Quadrature hybrid or90° hybrid couplers.With all the ports matched, power entering port 1 is eventually divided between ports 2 and 3 with a 90° phase shift between these outputs.No power is coupled to port 4.Branch line coupler has a high degree of symmetry, as any port can be used as the input port.The output ports will always be on the opposite of the junction from the input port and the isolated port will be the remaining port on the same side as the input port.The S-matrix will have the following form

$$[S] = \frac{-1}{\sqrt{2}} \begin{pmatrix} 0 & j & 1 & 0 \\ j & 0 & 0 & 1 \\ 1 & 0 & 0 & j \\ 0 & 1 & j & 0 \end{pmatrix}$$


Circuit Diagram:
L=10.6717mm
W=7.96336mm
P1
P2
Z=50Ω
Z=50Ω
W=4.86049mm
L=10.8525mm
L=10.8525mm
W=4.86049mm
P3
P4
Z=50Ω
L=10.6717mm
W=7.96336mm
Z=50Ω

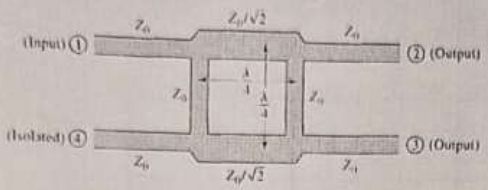


Figure (2)

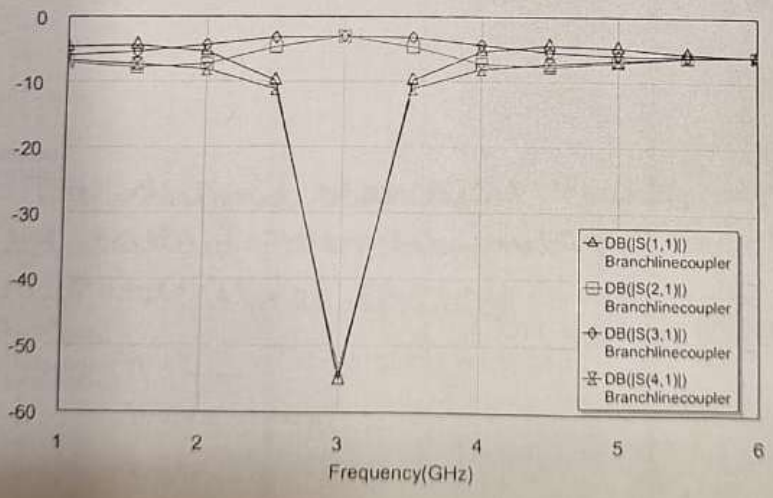
Sample Observations:

Z_0=50Ω f=3GHz, H=1.6mm, T=0.036 mm, ε_r=4.4, L=0.001

Table: S-Parameter In dB

Frequency (GHz)	DB(|S(1,1)|) Branch line coupler	DB(|S(2,1)|) Branch line coupler	DB(|S(3,1)|) Branch line coupler	DB(|S(4,1)|) Branch line coupler
1	-4.675	-7.1169	-5.9685	-6.7726
1.5	-4.3966	-7.8317	-5.4462	-7.3513
2	-5.3863	-7.3076	-4.3473	-8.1426
2.5	-9.8924	-4.6574	-3.2477	-11.203
3	-55.178	-3.0444	-3.0424	-53.669
3.5	-9.861	-4.6772	-3.2573	-11.18
4	-5.3872	-7.3093	-4.3794	-8.1248
4.5	-4.4449	-7.7704	-5.4826	-7.3198
5	-4.784	-7.0272	-5.9934	-6.724
5.5	-5.6372	-6.2806	-6.0754	-6.2163
6	-5.9985	-6.0597	-6.0649	-6.0654

Model graph: S-Parameter Plot



Practical Observations:

Frequency Mhz/Ghz	S_{11}	S_{21}	S_{31}	S_{41}
4.85	-24.844	-3.0807	-24.86	-3.023
4.87	-26.097	-3.0666	-26.091	-3.0228
4.89	-29.561	-3.0545	-27.532	-3.0226
4.9	-28.396	-3.0493	-28.355	-3.0225
4.93	-31.529	-3.0365	-31.438	-3.0223
4.96	-36.462	-3.0282	-36.276	-3.0222
5	-64.4	-3.0243	-60.312	-3.022
5.08	-30.197	-3.041	-30.298	-3.0218
5.12	-26.678	-3.0616	-26.791	-3.0218
5.18	-23.164	-3.1075	-23.301	-3.0222
5.23	-21.031	-3.1596	-21.209	-2.0232
5.28	-19.314	-3.224	-19.546	-3.0253

Conclusions:

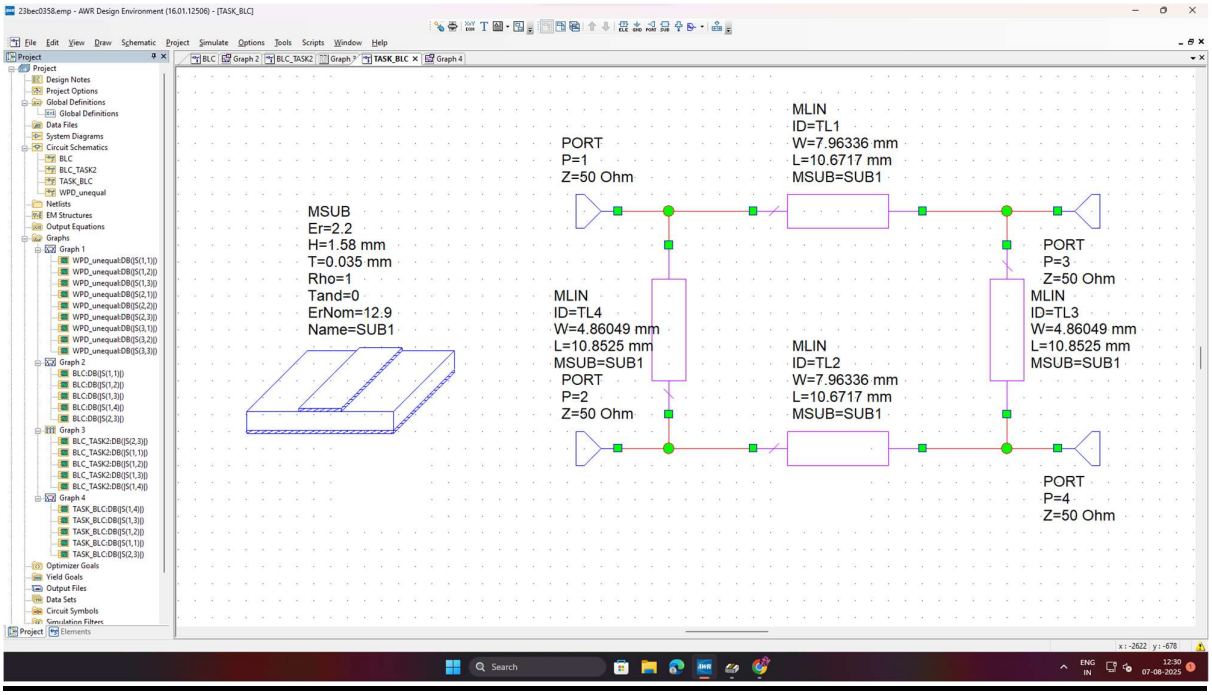
The designed branchline coupler meets the expected scattered Parameters, making it suitable for Quadrature signal splitting combines in Microwave application.

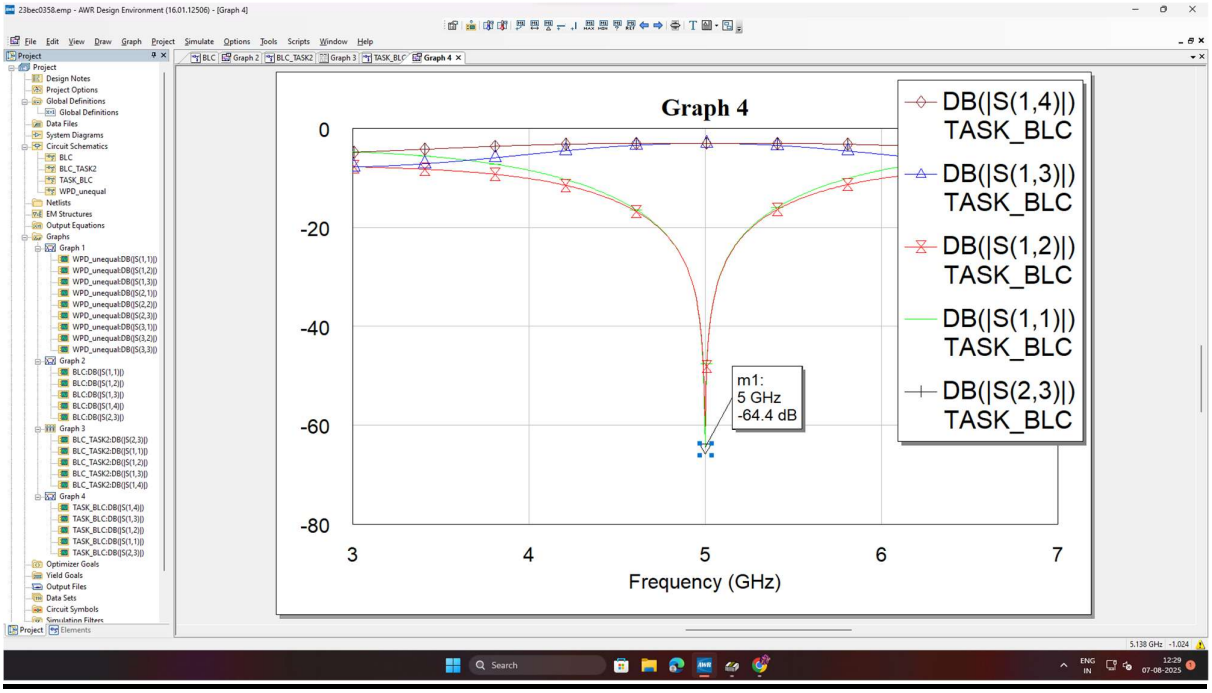
Evaluations: Viva: ____

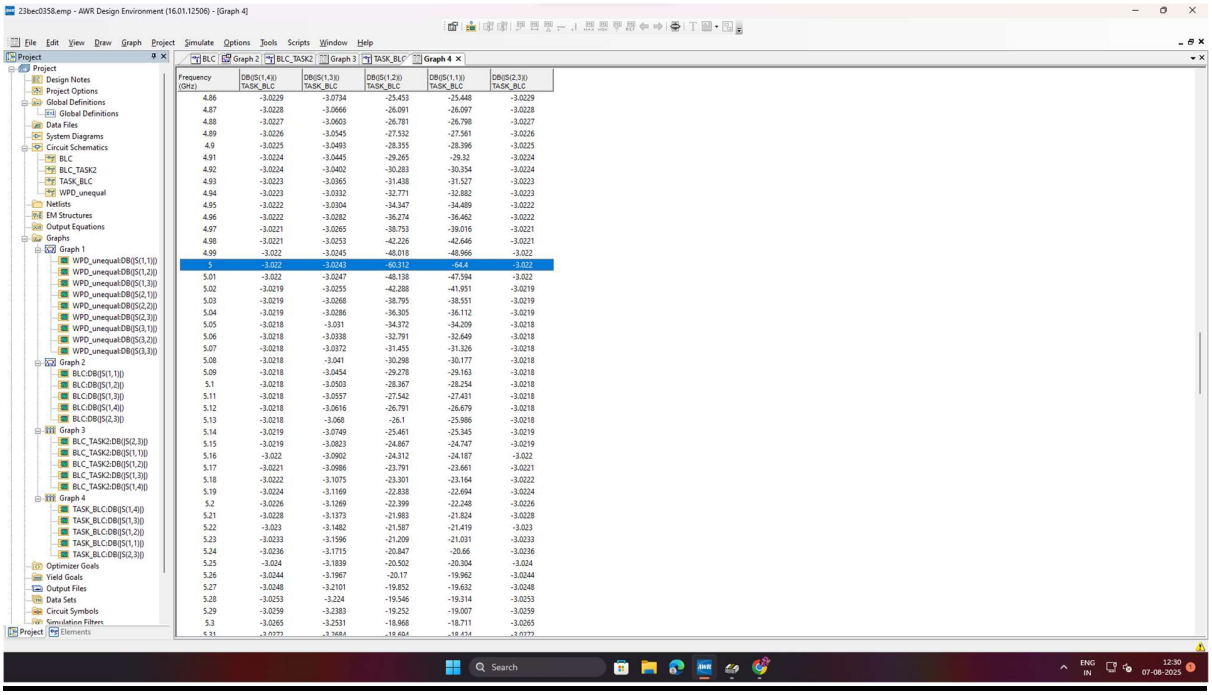
Design & Schematic:____

Results: ____

Faculty Signature


MSUB
Er=2.2
H=1.58 mm
T=0.035 mm
Rho=1
Tand=0
ErNom=12.9
Name=SUB1
PORT
P=1
Z=50 Ohm
MLIN
ID=TL1
W=7.96336 mm
L=10.6717 mm
MSUB=SUB1
PORT
P=3
Z=50 Ohm
MLIN
ID=TL4
W=4.86049 mm
L=10.8525 mm
MSUB=SUB1
MLIN
ID=TL3
W=4.86049 mm
L=10.8525 mm
MSUB=SUB1
MLIN
ID=TL2
W=7.96336 mm
L=10.6717 mm
MSUB=SUB1
PORT
P=2
Z=50 Ohm
PORT
P=4
Z=50 Ohm


Graph 4
0
-20
-40
-60
-80
3
4
5
6
7
Frequency (GHz)
m1:
5 GHz
-64.4 dB
DB(|S(1,4)|)
TASK_BLC
DB(|S(1,3)|)
TASK_BLC
DB(|S(1,2)|)
TASK_BLC
DB(|S(1,1)|)
TASK_BLC
DB(|S(2,3)|)
TASK_BLC



Frequency (GHz)	DB(|S(1,4)|) TASK_BLC	DB(|S(1,3)|) TASK_BLC	DB(|S(1,2)|) TASK_BLC	DB(|S(1,1)|) TASK_BLC	DB(|S(2,3)|) TASK_BLC
4.86	-3.0229	-3.0734	-25.453	-25.448	-3.0229
4.87	-3.0228	-3.0666	-26.091	-26.097	-3.0228
4.88	-3.0227	-3.0603	-26.781	-26.798	-3.0227
4.89	-3.0226	-3.0545	-27.532	-27.561	-3.0226
4.9	-3.0225	-3.0493	-28.355	-28.396	-3.0225
4.91	-3.0224	-3.0445	-29.265	-29.32	-3.0224
4.92	-3.0224	-3.0402	-30.283	-30.354	-3.0224
4.93	-3.0223	-3.0365	-31.438	-31.527	-3.0223
4.94	-3.0223	-3.0332	-32.771	-32.882	-3.0223
4.95	-3.0222	-3.0304	-34.347	-34.489	-3.0222
4.96	-3.0222	-3.0282	-36.274	-36.462	-3.0222
4.97	-3.0221	-3.0265	-38.753	-39.016	-3.0221
4.98	-3.0221	-3.0253	-42.226	-42.646	-3.0221
4.99	-3.022	-3.0245	-48.018	-48.966	-3.022
5	-3.022	-3.0243	-60.312	-64.4	-3.022
5.01	-3.022	-3.0247	-48.138	-47.594	-3.022
5.02	-3.0219	-3.0255	-42.288	-41.951	-3.0219
5.03	-3.0219	-3.0268	-38.795	-38.551	-3.0219
5.04	-3.0219	-3.0286	-36.305	-36.112	-3.0219
5.05	-3.0218	-3.031	-34.372	-34.209	-3.0218
5.06	-3.0218	-3.0338	-32.791	-32.649	-3.0218
5.07	-3.0218	-3.0372	-31.455	-31.326	-3.0218
5.08	-3.0218	-3.041	-30.298	-30.177	-3.0218
5.09	-3.0218	-3.0454	-29.278	-29.163	-3.0218
5.1	-3.0218	-3.0503	-28.367	-28.254	-3.0218
5.11	-3.0218	-3.0557	-27.542	-27.431	-3.0218
5.12	-3.0218	-3.0616	-26.791	-26.679	-3.0218
5.13	-3.0218	-3.068	-26.1	-25.986	-3.0218
5.14	-3.0219	-3.0749	-25.461	-25.345	-3.0219
5.15	-3.0219	-3.0823	-24.867	-24.747	-3.0219
5.16	-3.022	-3.0902	-24.312	-24.187	-3.022
5.17	-3.0221	-3.0986	-23.791	-23.661	-3.0221
5.18	-3.0222	-3.1075	-23.301	-23.164	-3.0222
5.19	-3.0224	-3.1169	-22.838	-22.694	-3.0224
5.2	-3.0226	-3.1269	-22.399	-22.248	-3.0226
5.21	-3.0228	-3.1373	-21.983	-21.824	-3.0228
5.22	-3.023	-3.1482	-21.587	-21.419	-3.023
5.23	-3.0233	-3.1596	-21.209	-21.031	-3.0233
5.24	-3.0236	-3.1715	-20.847	-20.66	-3.0236
5.25	-3.024	-3.1839	-20.502	-20.304	-3.024
5.26	-3.0244	-3.1967	-20.17	-19.962	-3.0244
5.27	-3.0248	-3.2101	-19.852	-19.632	-3.0248
5.28	-3.0253	-3.224	-19.546	-19.314	-3.0253
5.29	-3.0259	-3.2383	-19.252	-19.007	-3.0259
5.3	-3.0265	-3.2531	-18.968	-18.711	-3.0265

Expt. No.: Task-3

Date : 7/8/25

180^0 Hybrid Coupler

Objectives:

5. To design and simulation of 180^0 hybrid coupler.
6. To determine the scattering parameters of 180^0 Hybridcoupler.

Equipment required: AWR Microwave Office software

Specifications:

Characteristic impedance	Z_0= 50Ω
Operating frequency	f = 3.56 GHz
Substrate thickness	H= 1.58 mm
Dielectric constant	ε_r= 2.2
Loss tangent	L= 0.02.

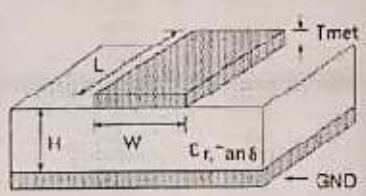


Figure (1) Microstrip Line

Theory : The 180^0 hybrid junction is a four port networ with a 180^0 phase shift between the two output ports.It can also be operated so that the outputs are in phase.With reference to the hybrid coupler shown in figure 2, a signal applied to port 1 will be evenly split into two in-phase components at ports 2 and 3,and port 4 will be isolated.If the input is applied to port 4, it will be equallysplit into two compoents with a 180^0 phase difference at ports 2 and 3, andport 1 will be isolated.When operated as a combiner with input signals applied at ports 2 and 3, sum of theinputs will be formed at port 1, whilethe difference will be formed at port 4. The 180 hybrid can be fabricated in several forms.The ring hybrid or rat-race shown in figure 2 can be easily constructed in microstrip formshown in figure 2.Thescattering matrixfor theideal 3 dB180^{0}hybridthushas the following form

$$[S] = \frac{-j}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \end{pmatrix}$$

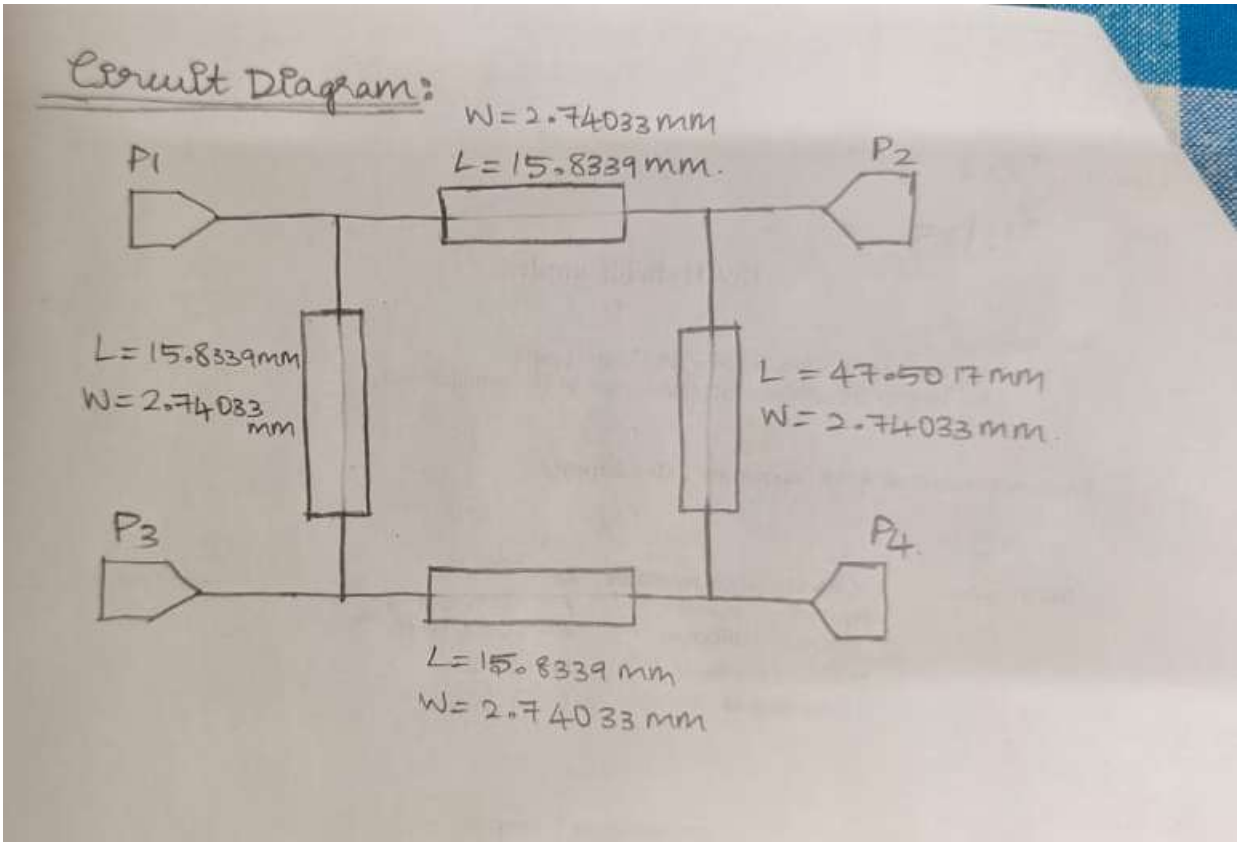
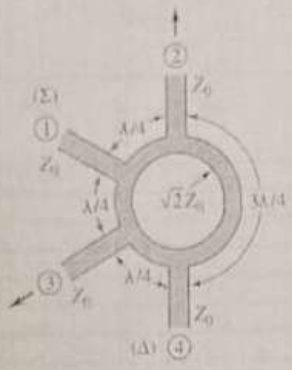

Circuit Diagram:
W=2.74033mm
L=15.8339mm.
P1
P2
L=15.8339mm
W=2.74033 mm
L=47.5017mm
W=2.74033mm.
P3
P4
L=15.8339 mm
W=2.74033 mm

Figure (2)



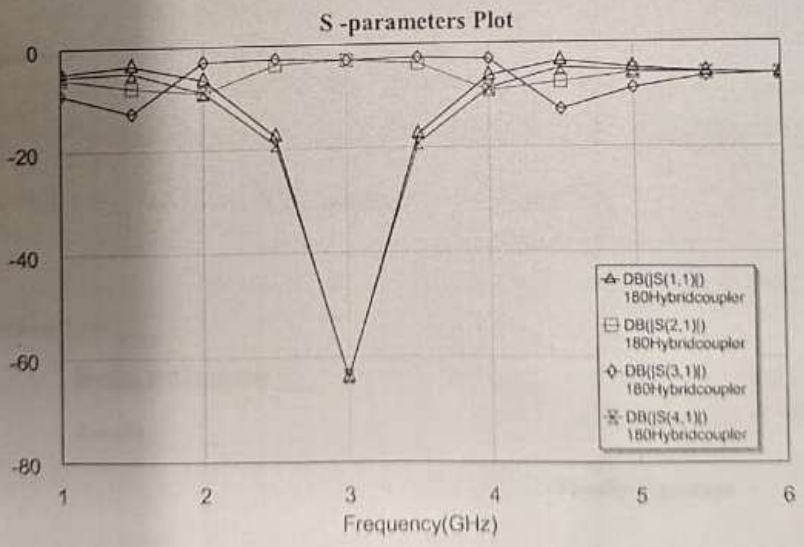
Sample Observations:

Z_0=50Ω, f=3GHz, H=1.6mm, T=0.036 mm, ε_r=4.4, L=0.001

Table: S-Parameter In dB

Frequency (GHz)	DB(|S(1,1)|) 180 Hybrid coupler	DB(|S(2,1)|) 180 Hybrid coupler	DB(|S(3,1)|) 180 Hybrid coupler	DB(|S(4,1)|) 180 Hybrid coupler
1	-4.8376	-5.9388	-9.0488	-5.3951
1.5	-3.496	-7.7967	-12.702	-4.8726
2	-6.4332	-9.0219	-2.9521	-8.841
2.5	-17.604	-3.7851	-2.6307	-19.229
3	-63.9	-3.0438	-3.0627	-63.815
3.5	-17.49	-3.8045	-2.6352	-19.181
4	-6.3076	-9.2147	-3.0407	-8.7297
4.5	-3.5817	-7.5942	-12.865	-4.9205
5	-4.9584	-5.9166	-8.7864	-5.483
5.5	-5.8106	-5.9885	-6.51	-5.9542
6	-6.0375	-6.0462	-6.0705	-6.0819

Model graph:



Practical Observations:

Frequency Mhz/Ghz	S_{11}	S_{21}	S_{31}	S_{41}
3.17	-24.442	-3.2308	-2.8817	-25.185
3.19	-25.101	-3.2054	-2.8976	-25.778
3.23	-26.512	-3.1603	-2.9269	-27.07
3.28	-28.51	-3.114	-2.9581	-28.944
3.3	-29.41	-3.0984	-2.969	-29.805
3.34	-31.46	-3.0721	-2.9876	-31.8
3.38	-34.016	-3.0519	-3.0022	-34.346
3.45	-41.421	-3.0309	-3.0179	-42.024
3.5	-62.981	-3.0066	-3.0012	-69.946
3.52	-51.27	-3.0274	-3.0207	-49.743
3.54	-44.575	-3.0296	-3.0191	-43.824
3.57	-39.381	-3.0356	-3.0148	-38.99

Conclusions:

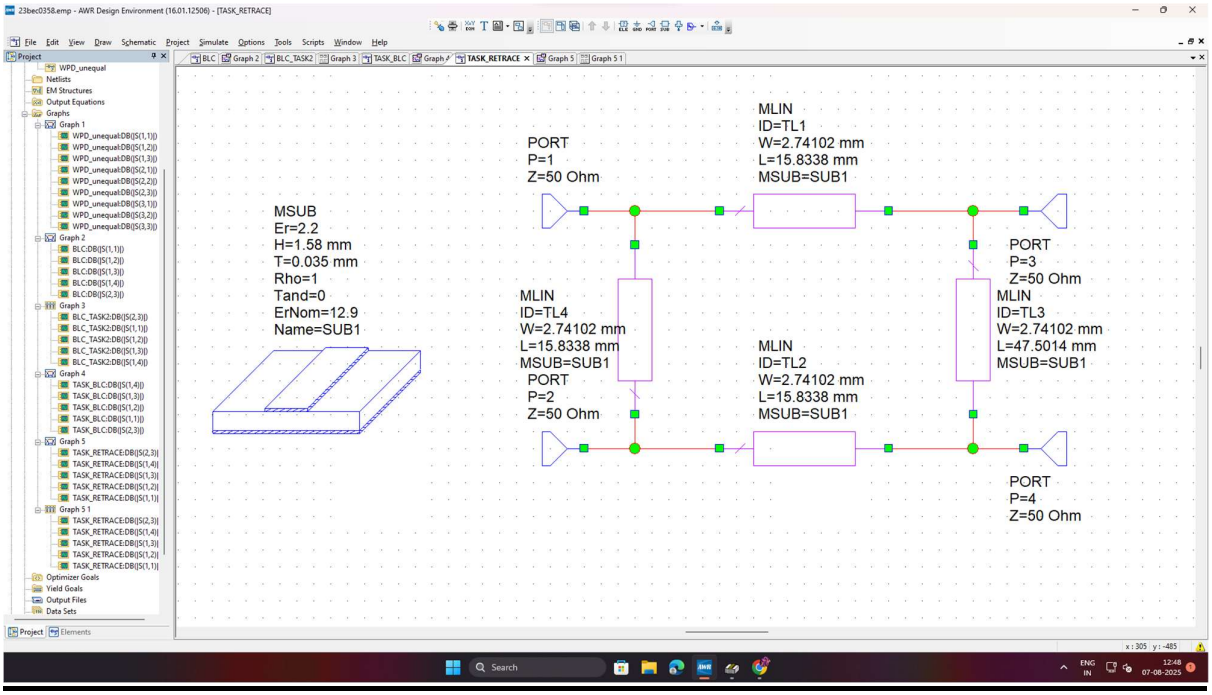
The 180° hybrid coupler meets the expected scattering parameters & can be used for Power Division (or) Combination in Microwave Circuits.

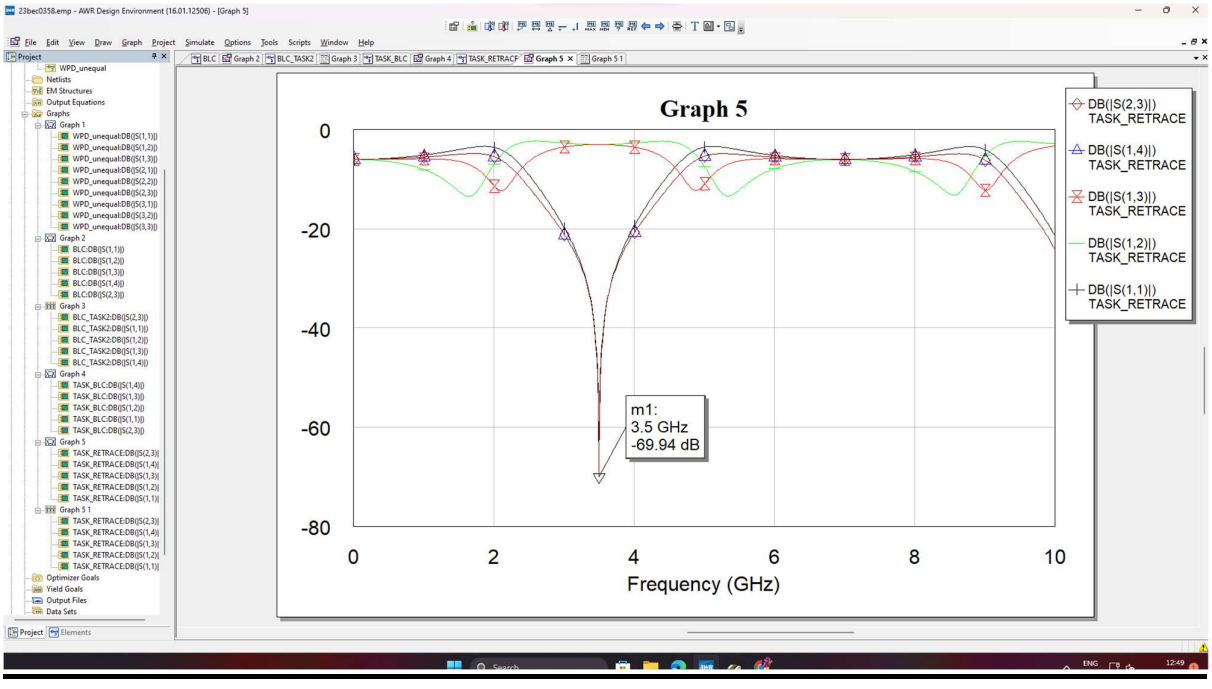
Evaluations: Viva ____

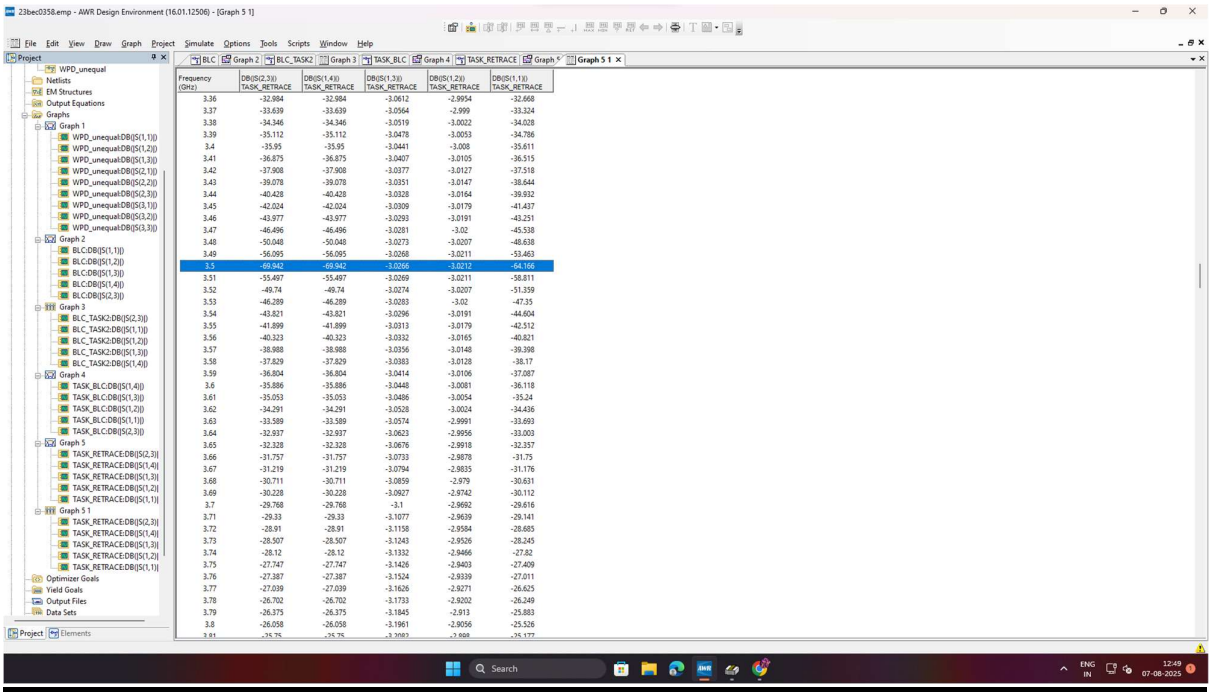
Design & Schematic ____

Results _____

Faculty Signature


MSUB
Er=2.2
H=1.58 mm
T=0.035 mm
Rho=1
Tand=0
ErNom=12.9
Name=SUB1
PORT
P=1
Z=50 Ohm
MLIN
ID=TL1
W=2.74102 mm
L=15.8338 mm
MSUB=SUB1
PORT
P=3
Z=50 Ohm
MLIN
ID=TL4
W=2.74102 mm
L=15.8338 mm
MSUB=SUB1
MLIN
ID=TL3
W=2.74102 mm
L=47.5014 mm
MSUB=SUB1
MLIN
ID=TL2
W=2.74102 mm
L=15.8338 mm
MSUB=SUB1
PORT
P=2
Z=50 Ohm
PORT
P=4
Z=50 Ohm


Graph 5
0
-20
-40
-60
-80
0
2
4
6
8
10
Frequency (GHz)
m1:
3.5 GHz
-69.94 dB
DB(|S(2,3)|)
TASK_RETRACE
DB(|S(1,4)|)
TASK_RETRACE
DB(|S(1,3)|)
TASK_RETRACE
DB(|S(1,2)|)
TASK_RETRACE
DB(|S(1,1)|)
TASK_RETRACE


23bec0358.emp - AWR Design Environment (16.01.12506) - [Graph 5 1]
File Edit View Draw Graph Project Simulate Options Tools Scripts Window Help
Project
WPD_unequal
Netlists
EM Structures
Output Equations
Graphs
Graph 1
WPD_unequal:DB(|S(1,1)|)
WPD_unequal:DB(|S(1,2)|)
WPD_unequal:DB(|S(1,3)|)
WPD_unequal:DB(|S(2,1)|)
WPD_unequal:DB(|S(2,2)|)
WPD_unequal:DB(|S(2,3)|)
WPD_unequal:DB(|S(3,1)|)
WPD_unequal:DB(|S(3,2)|)
WPD_unequal:DB(|S(3,3)|)
Graph 2
BLC:DB(|S(1,1)|)
BLC:DB(|S(1,2)|)
BLC:DB(|S(1,3)|)
BLC:DB(|S(1,4)|)
BLC:DB(|S(2,3)|)
Graph 3
BLC_TASK2:DB(|S(2,3)|)
BLC_TASK2:DB(|S(1,1)|)
BLC_TASK2:DB(|S(1,2)|)
BLC_TASK2:DB(|S(1,3)|)
BLC_TASK2:DB(|S(1,4)|)
Graph 4
TASK_BLC:DB(|S(1,4)|)
TASK_BLC:DB(|S(1,3)|)
TASK_BLC:DB(|S(1,2)|)
TASK_BLC:DB(|S(1,1)|)
TASK_BLC:DB(|S(2,3)|)
Graph 5
TASK_RETRACE:DB(|S(2,3)|)
TASK_RETRACE:DB(|S(1,4)|)
TASK_RETRACE:DB(|S(1,3)|)
TASK_RETRACE:DB(|S(1,2)|)
TASK_RETRACE:DB(|S(1,1)|)
Graph 5 1
TASK_RETRACE:DB(|S(2,3)|)
TASK_RETRACE:DB(|S(1,4)|)
TASK_RETRACE:DB(|S(1,3)|)
TASK_RETRACE:DB(|S(1,2)|)
TASK_RETRACE:DB(|S(1,1)|)
Optimizer Goals
Yield Goals
Output Files
Data Sets
Project Elements
BLC Graph 2 BLC_TASK2 Graph 3 TASK_BLC Graph 4 TASK_RETRACE Graph 5 Graph 5 1
Frequency (GHz)
DB(|S(2,3)|) TASK_RETRACE
DB(|S(1,4)|) TASK_RETRACE
DB(|S(1,3)|) TASK_RETRACE
DB(|S(1,2)|) TASK_RETRACE
DB(|S(1,1)|) TASK_RETRACE
3.36 -32.984 -32.984 -3.0612 -2.9954 -32.668
3.37 -33.639 -33.639 -3.0564 -2.999 -33.324
3.38 -34.346 -34.346 -3.0519 -3.0022 -34.028
3.39 -35.112 -35.112 -3.0478 -3.0053 -34.786
3.4 -35.95 -35.95 -3.0441 -3.008 -35.611
3.41 -36.875 -36.875 -3.0407 -3.0105 -36.515
3.42 -37.908 -37.908 -3.0377 -3.0127 -37.518
3.43 -39.078 -39.078 -3.0351 -3.0147 -38.644
3.44 -40.428 -40.428 -3.0328 -3.0164 -39.932
3.45 -42.024 -42.024 -3.0309 -3.0179 -41.437
3.46 -43.977 -43.977 -3.0293 -3.0191 -43.251
3.47 -46.496 -46.496 -3.0281 -3.02 -45.538
3.48 -50.048 -50.048 -3.0273 -3.0207 -48.638
3.49 -56.095 -56.095 -3.0268 -3.0211 -53.463
3.5 -69.942 -69.942 -3.0266 -3.0212 -64.166
3.51 -55.497 -55.497 -3.0269 -3.0211 -56.811
3.52 -49.74 -49.74 -3.0274 -3.0207 -51.359
3.53 -46.289 -46.289 -3.0283 -3.02 -47.35
3.54 -43.821 -43.821 -3.0296 -3.0191 -44.604
3.55 -41.899 -41.899 -3.0313 -3.0179 -42.512
3.56 -40.323 -40.323 -3.0332 -3.0165 -40.821
3.57 -38.988 -38.988 -3.0356 -3.0148 -39.398
3.58 -37.829 -37.829 -3.0383 -3.0128 -38.17
3.59 -36.804 -36.804 -3.0414 -3.0106 -37.087
3.6 -35.886 -35.886 -3.0448 -3.0081 -36.118
3.61 -35.053 -35.053 -3.0486 -3.0054 -35.24
3.62 -34.291 -34.291 -3.0528 -3.0024 -34.436
3.63 -33.589 -33.589 -3.0574 -2.9991 -33.693
3.64 -32.937 -32.937 -3.0623 -2.9956 -33.003
3.65 -32.328 -32.328 -3.0676 -2.9918 -32.357
3.66 -31.757 -31.757 -3.0733 -2.9878 -31.75
3.67 -31.219 -31.219 -3.0794 -2.9835 -31.176
3.68 -30.711 -30.711 -3.0859 -2.979 -30.631
3.69 -30.228 -30.228 -3.0927 -2.9742 -30.112
3.7 -29.768 -29.768 -3.1 -2.9692 -29.616
3.71 -29.33 -29.33 -3.1077 -2.9639 -29.141
3.72 -28.91 -28.91 -3.1158 -2.9584 -28.685
3.73 -28.507 -28.507 -3.1243 -2.9526 -28.245
3.74 -28.12 -28.12 -3.1332 -2.9466 -27.82
3.75 -27.747 -27.747 -3.1426 -2.9403 -27.409
3.76 -27.387 -27.387 -3.1524 -2.9339 -27.011
3.77 -27.039 -27.039 -3.1626 -2.9271 -26.625
3.78 -26.702 -26.702 -3.1733 -2.9202 -26.249
3.79 -26.375 -26.375 -3.1845 -2.913 -25.883
3.8 -26.058 -26.058 -3.1961 -2.9056 -25.526
Search
ENG IN
12:49 07-08-2025